# Network Contagion, Heavy-Tail Dynamics and Default Cascades

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## Introduction

Financial markets exhibit complex interdependencies where local shocks may propagate through network structures, generating systemic risk.

This study develops a dynamic contagion framework integrating:

- Correlation-based financial networks
- Tail risk measures (VaR, CVaR)
- Heavy-tail behavior via Hill estimator
- Threshold-based default cascades

The objective is to analyze how structural dependencies and extreme events jointly drive systemic instability. This approach extends previous work [2] by incorporating heavy-tail behavior and network-driven contagion.

## Model Framework

Let $r_i(t)$ denote the log-return of asset $i$.

The network is defined by a weighted adjacency matrix:

$$W_{ij} = \frac{|\rho_{ij}|}{\sum_k |\rho_{ik}|}$$

where $\rho_{ij}$ is the correlation between assets.

Edges are filtered using a threshold $\rho_{min}$ and top-$k$ selection.

## Risk Measures

Define losses:

$$L_i = -r_i$$

Value-at-Risk:

$$VaR_i = \inf\{x : P(L_i \leq x) \geq 1 - \alpha\}$$

Conditional Value-at-Risk:

$$CVaR_i = E[L_i \mid L_i \geq VaR_i]$$

## Default Cascade Dynamics

Let $D_j \in \{0, 1\}$ indicate default state.

The incoming shock to node $i$ is:

$$S_i = \sum_j W_{ij} \cdot Shock_j \cdot D_j$$

Default condition:

$$\frac{S_i}{C_i} > \theta$$

where $C_i$ represents capital (volatility-based).

## Heavy-Tail Analysis

Tail behavior is estimated using the Hill estimator:

$$\alpha^{-1} = \frac{1}{k} \sum_{i=1}^{k} \log\left(\frac{X_i}{X_{k+1}}\right)$$

Assets with $\alpha < 3$ exhibit heavy-tail risk.

## Network Interpretation

The system reveals:

- Clustering of highly correlated assets
- Shock amplification through dense subgraphs
- Asymmetry between emerging and developed markets

Brazilian assets exhibit stronger clustering, while global assets act as shock transmitters.

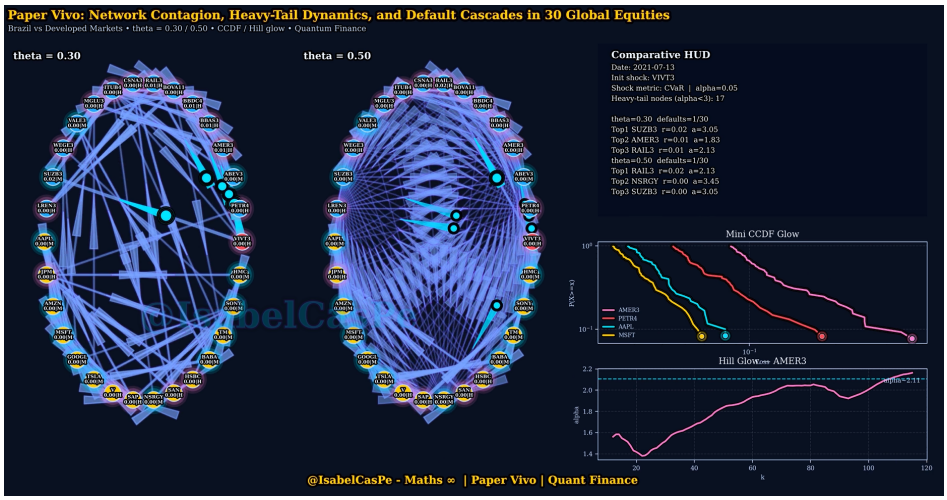


Figure 1: **Snapshot of the V.11 contagion dynamics.** This frame highlights the network-based transmission of shocks across assets, emphasizing cascade formation, clustering, and heavy-tail amplification effects in the systemic risk model.

As shown in Figure 1, the contagion process emerges from the interaction between network connectivity, threshold effects, and tail-driven stress propagation.

# Results and Visual Interpretation

The animation (V.11) illustrates:

- Dynamic propagation of shocks
- Tail-driven contagion amplification
- Phase transition behavior under different $\theta$

The system demonstrates resilience under moderate shocks but exhibits nonlinear amplification under heavy-tail conditions.

# Conclusion

Systemic risk emerges from the interaction between:

- Network topology
- Tail risk behavior
- Shock propagation mechanisms

This framework provides a unified approach to understanding financial contagion beyond traditional risk measures.

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References

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